



Voting systems

Starter quiz

© Oak National Academy 2025. Produced in partnership with Mathematics in Education and Industry.

Licensed on the [Open Government Licence v3.0](#), except where otherwise stated. [See Oak terms & conditions](#).



0

Voting systems

- 1 Here is a formula. If the values for a, b, c and d are known, what is a valid first step? (Tick **1** correct answer)

$$\frac{a^2 + b}{c - d} = h$$

- ☐ Evaluate the numerator.
- ☐ Evaluate the denominator.
- ☐ Divide the numerator by the denominator.
- ☐ Substitute in the known values.

- 2 Here is a formula. If the values for a, b, c and d are known and substituted in, what is a valid next step? (Tick **2** correct answers)

$$\frac{a^2 + b}{c - d} = h$$

- ☐ Evaluate the numerator.
- ☐ Evaluate the denominator.
- ☐ Divide the numerator by the denominator.
- ☐ Substitute in the known values.

- 3 Here is a formula. If $a = 5$, $b = 7$, $c = 19$ and $d = 11$ what does the numerator evaluate to?

$$\frac{a^2 + b}{c - d} = h$$

- 4 Here is a formula. If $a = 5$, $b = 7$, $c = 19$ and $d = 11$ what does the denominator evaluate to?

$$\frac{a^2 + b}{c - d} = h$$

- 5 Here is a formula. If $a = 5$, $b = 7$, $c = 19$ and $d = 11$ which of the following is correct? (Tick **1** correct answer)

$$\frac{a^2 + b}{c - d} = h$$

☐ $h = \frac{32}{9}$

☐ $h = \frac{52}{9}$

☐ $h = \frac{32}{8}$

☐ $h = \frac{52}{8}$

6 Here is a formula. If $a = 5$, $b = 7$, $c = 19$ and $d = 11$ what is the value of h ?

$$\frac{a^2 + b}{c - d} = h$$
